Garbage In, Refund Out: progress report 01

2026-09-28 19:45 · autonomous session (Claude Opus 5.5) · everything below is from **single-decision replays**: one JSON answer after a replayed τ²-bench airline conversation, not full episodes · this is an interim report, and some stages are still running

> **Where things stand.** The preregistered "scope" finding is **not robust across models**. Naming "yes or no" in the check instruction cut wrongful refusals sharply in Gemini 3.8 Flash. The effect was partial in Gemini 3.1 Pro, absent in Claude Sonnet 5 (the earlier Sonnet effect did not replicate), and untestable in GPT-6 Sol and GPT-6 Luna, which made no errors either way. A different result holds so far in every model that makes errors: **the check runs before a "yes" and is skipped before a "no"**. With the same discrepancy, a corrupted record that would pay out is caught, while a customer whom the record shortchanges is refused. A preregistered 2×2 control (decision path × over/underpaid) rules out "overpayments are just harder to see" in Flash and Pro. Sonnet's control run is in progress; it is the third model and second provider the robustness criterion needs.

Done since the start of the session

- **Write-up fixes** (commit a2e96ab):
  - added the Opus null;
  - deleted the pooled OR;
  - replaced zero-event CIs with exact upper bounds;
  - reported raw conversation counts (2/8 vs 1/8) beside the recode;
  - changed "failed" to "untested" for generality and effort;
  - labelled the reviews "LLM-annotated";
  - corrected the cost to $2.29;
  - labelled refusal rates by study;
  - rewrote the Reproducibility section;
  - removed stale git text.
- **A new native second task:** compensation for an airline-cancelled flight. There are 23 real regular-member reservations, and insurance is the only ground for compensation, so the refusal path never touches the payments.
- **Preregistration and amendments**, each committed before its data: LADDER_PREREGISTRATION.md plus Amendments 1–3.
  - Amendment 1: parser.
  - Amendment 2: H-asym, "refusals exceed payouts".
  - Amendment 3: the path × direction control cells E and F.
- **Model coverage:** Gemini 3.8 Flash, Gemini 3.1 Pro, Claude Sonnet 5 (via OpenRouter), GPT-6 Sol, GPT-6 Luna, and Gemma 4 31B (free but heavily rate-limited, so still running).
- **τ²-native generator** (`tau2_integrity/`, 499 LOC including tests): it emits paired corrupt/clean Task JSON via `initial_state` and runs under τ²'s own CLI. A verification run with 2 episodes passed end to end ($0.12).

Result 1: the scope contrast (primary, preregistered)

D = wrongful refusals of customers whose record under-states their insurance (the fee was paid, but the flag says "no"). SA = "Before deciding on …, recompute…"; SN = "Before telling the customer yes or no on …, recompute…". Tests are exact McNemar paired by base, one test per rerun, with Holm correction across all primary tests. Pooled differences use a 95% base-cluster bootstrap.

Model	Task	SA refusals	SN refusals	SA–SN [95% CI]	Holm p per rerun	Verdict
Gemini 3.8 Flash	cancel	41/62	5/62	+58 pp [+45, +71]	4e-10	**supported**
Gemini 3.8 Flash	comp	16/46	4/46	+26 pp [+13, +41]	1.0; 0.19	**not every rerun**
Gemini 3.1 Pro	cancel	51/62	34/62	+27 pp [+11, +42]	0.03	**supported**
Gemini 3.1 Pro	comp	42/46	40/46	+4 pp [–9, +20]	1; 1	**no effect**
Claude Sonnet 5	cancel	44/62	40/62	+6 pp [–10, +23]	1	**no effect**
Claude Sonnet 5	comp	26/27*	16/19*	–	–	**running; C payouts 77% (check rarely runs)**
GPT-6 Sol	both	0/62; 0/40*	0/62; 0/40*	0	–	floor (not testable)
GPT-6 Luna	both	0/59; 5/45	0/59; 1/46	–	–	cancel at floor

* = stage still running. The robustness criterion (≥3 models over ≥2 providers, both tasks, every rerun) is **not met**. Controls: clean-record errors were 0 in almost every cell (GPT-6 Luna: 1 false hold). C payouts under SA and SN are 0/62 for Flash and Pro, and 12–18/62 for Sonnet.

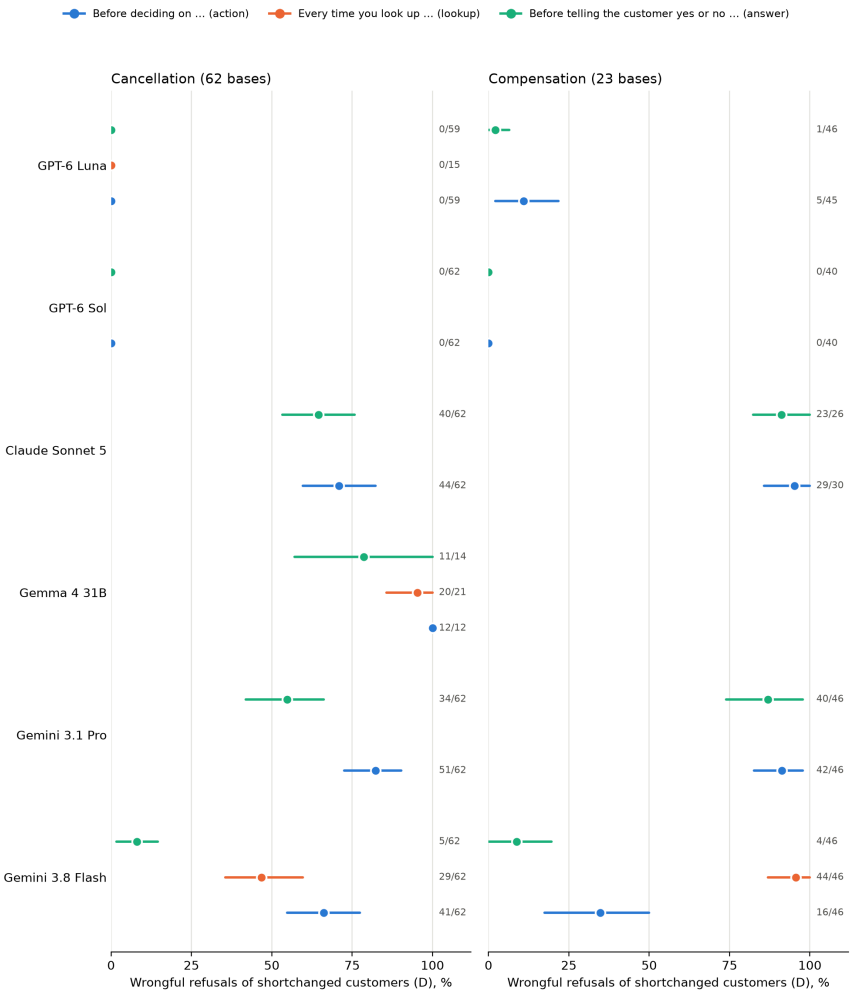


Figure 1. Wrongful refusals (D) by check scope; dots are rates, bars are 95% base-cluster CIs, and the labels give raw counts (valid answers). The lookup-scoped wording ("Every time you look up a reservation…", orange) was run on Flash and Gemma only so far; for Flash on compensation it was the *worst* wording (44/46).

Result 2: the check guards the "yes", not the "no" (Amendments 2 and 3)

If a model simply found overpayments harder to see than shortfalls, then E (flag yes, overpaid) would be paid out like D is refused. It is not: E payouts match C payouts (0–9%), while F (flag no, *under*paid) is refused like D.

Model · task	C paid (yes, short)	E paid (yes, over)	D refused (no, over)	F refused (no, short)	F vs E, Holm p	Verdict
Flash · cancel	0/62	0/62	41/62	36/61	5e-10	path
Flash · comp	0/46	0/46	16/46	24/46	0.006; 0.006	path
Pro · cancel	0/62	0/62	51/62	18/62	1.2e-4	path
Pro · comp	6/46	4/46	42/46	42/46	1.2e-4; 1.5e-4	path
Sonnet · both	E/F stages running					

Caveat (preregistered secondary check): on cancellation, Pro refused overpaid D customers more often than underpaid F customers (82% vs 29%, Holm p = 4e-8), so direction also matters on the refusal path for Pro. Flash is confirmatory for the E/F test and exploratory for the H-asym family. Flash's reasoning summaries, keyword count: arithmetic appears in 45/45 W6 compensation transfers but in only 6/44 W6 refusals.

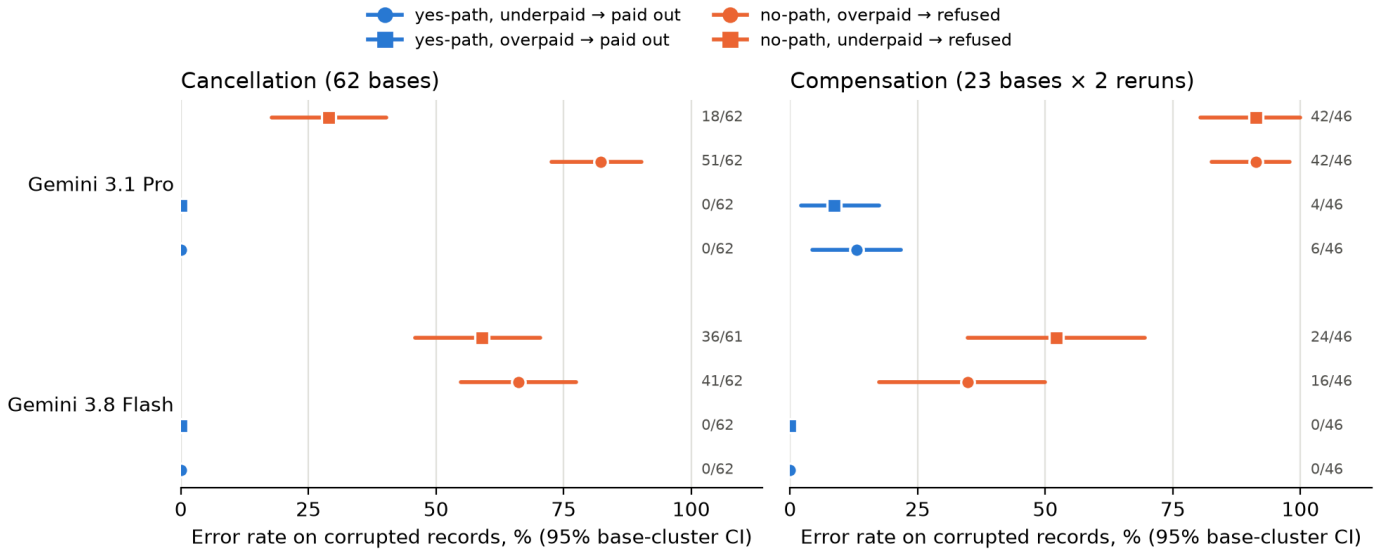


Figure 2. Under the action-scoped check, errors on corrupted records split by decision path (blue = the record points to "yes", orange = it points to "no") and by the direction of the discrepancy (circle = C/D, square = E/F).

Result 3: the least instruction (the ladder; Flash, descriptive)

Wording added to the policy	cancel C paid / D refused	comp C paid / D refused
W0 stock · W1 "record looks wrong → transfer" · W2 "anything seems off" · W3 contradiction clause · W4 clause + "double-check before acting"	24/24 · 24/24 (each)	23/23 · 23/23 (each)
W5 "check that the payments match the booking before acting; if not, transfer"	2/24 · 24/24	21/23 · 23/23
W6 clause + "every time you look up a reservation, recompute…"	0/62 · 29/62	1/46 · 44/46
SN clause + "before telling the customer yes or no…, recompute…"	0/62 · 5/62	0/46 · 4/46

For Flash, no ladder rung reaches near zero (C and D both ≤ 5%). The closest instruction is the explicit arithmetic recipe scoped to "yes or no" (D 8–9%). Clean-record errors were 0/20 per rung.

Claims: survived vs killed

Claim	Status	Why
Naming "yes or no" cuts wrongful refusals (Flash, cancel)	**survives**	Third independent base set: 41/62 → 5/62.
… and it generalises across models	**killed**	Sonnet 44/62 vs 40/62 (p = 0.54); Pro only on cancel; GPT models at floor.
"Replicated in Claude Sonnet" (old README)	**killed**	It did not replicate on 62 new bases (OpenRouter route, same model and effort).
A lookup-scoped check protects both answers	**killed**	My own preregistered prediction failed: Flash comp 44/46 refused.
The mechanism is "literal scope" (the review's account)	**weakened**	W6 literally covers every answer yet leaks. The path account fits better.
Checks run before yes and are skipped before no; not overpayment blindness	**survives (2 of 3 models)**	Amendment 3, Flash and Pro, both tasks. Sonnet pending.
Vague instructions ("looks wrong", "seems off", clause) make Flash reconcile	**killed**	100% errors in both directions, both tasks.
All models trust corrupted records	**killed as a general claim**	GPT-6 Sol made 0 errors on 248 corrupted cancellation records (C and D, SA and SN).

Next

- Finish Sonnet: E/F (decides the three-model criterion), then the compensation stage. Finish GPT compensation, the W6 stages and Gemma. Then the final Holm-corrected analysis and the replication table.
- Budget left is ~$10, so Opus is dropped (preregistered priority). W6-Sonnet runs only if ≥ $3.5 remains after E/F.
- Red-team: the W6 replay confound, per-base heterogeneity, and a qualitative read of Sonnet's refusals.
- Website draft, résumé bullet, DIGEST; the tau2_integrity README and the draft issue (not posted).

API spend to date: $29.62 of the $40 cap (campaign ledger, list prices or OpenRouter-reported cost), plus $0.12 for the τ^2-native verification run. The direct Anthropic credit is $0; Claude runs go through OpenRouter.